

Derek Wang

+1 (416) 898-6127 | derek.wang1@uwaterloo.ca | [linkedin.com/in/4derekwang](https://www.linkedin.com/in/4derekwang) | github.com/4ppleSA0CE | derekwng.ca

EDUCATION

University of Waterloo | *Bachelor of Applied Science in Computer Engineering (Co-op)* Expected April 2030

EXPERIENCE

Software Engineer Intern May 2026 – August 2026
Prandtl Dynamics Inc. Toronto, ON

- Closed cross-tenant data exposure across 6 dashboard services with middleware that injects org-ID predicates into every query, scopes WebSocket rooms per org, and fails closed on JWT checks, enforced by CI audit gates
- Tracked 5+ simultaneous DJI drones at 170 m with 20 ms latency via a real-time Python pipeline chaining sound-source localization, beamforming, and Kalman-filter tracking on a single microphone array
- Raised acoustic drone detection from 85% to 98% mAP on a held-out split, with 40% fewer false alarms at half the size, by replacing the YAMNet audio classifier with EfficientAT in PyTorch
- Automated cleaning of 50+ hours of training audio with a Python pipeline that localizes and beamforms recordings

Software Engineer Intern March 2026 – June 2026
Kenosis Technologies Inc. Mississauga, ON

- Lowered API latency 35% by caching posting lookups and auth sessions in Redis across TypeScript/Node.js services
- Cut job-posting refresh time from daily to hourly by designing the MongoDB schemas and ingestion pipelines behind it
- Flagged suspicious applicants 60% faster by running LLM fraud screening and document review on async job queues
- Created candidate distance and map APIs with MongoDB geospatial queries to filter applicants by location

Autonomy Software Developer September 2025 – Present
WATonomous – Student Design Team Waterloo, ON

- Enabled 2 Hz obstacle-aware rover replanning via C++ ROS 2 A* on a 30x30 m LiDAR costmap and 10 Hz Pure Pursuit
- Led YOLO11s object detection retraining on 3,916 images to 0.96 mean average precision (mAP50) and accelerated inference to 30 FPS on the rover's NVIDIA Jetson Nano via ONNX export and onnxslim

PROJECTS

Reparo – AI Repair Assistant | Hack Canada 2026 Winner (\$5,000) | *Python, FastAPI, Swift, Gemini* [GitHub](#)

- Turned one device photo into a structured repair plan in under 60 s via component detection and Gemini reasoning
- Made every repair plan render in one format by normalizing raw Gemini output into typed FastAPI repair models
- Collapsed parts buying into one checkout with no app install via SerpAPI and Shopify APIs in a Swift App Clip

Mr. Clean – LLM-Orchestrated Mobile Manipulation | *Python, MuJoCo, PyTorch, Claude API* [GitHub](#) | [Demo](#)

- Drove a simulated two-arm robot across 3 tables from one typed prompt by building a Claude commander that hands each table to its own controller: an LLM skills agent, an imitation-learning policy, or a fly-connectome policy
- Sorted 4/4 cubes in 13 tool calls via a nested Claude agent whose 6 typed skills refuse unknown objects at the boundary
- Steered the base to 20/20 unseen goals with a policy wired from a fruit fly's brain (512-neuron FlyWire connectome)

7-DOF Robotic Arm Motion Control | *C++17, Eigen, MuJoCo, CMake, React, TypeScript* [GitHub](#)

- Wrote 7-DOF arm kinematics and dynamics from scratch in C++ using Lie groups and recursive Newton-Euler, matching MuJoCo on 1,000 test poses for forward kinematics and inverse dynamics
- Held tip error under 0.2 mm on 46/49 targets with zero actuator saturation via jerk-limited 1 kHz computed torque
- Removed arm vibration and a 2 cm tolerance by moving inverse kinematics out of the 1 kHz loop to replan on demand

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, Swift, SQL, MATLAB

Frameworks/Libraries: React, Next.js, Node.js, Express, FastAPI, PyTorch, OpenCV, ROS 2, ONNX

Infrastructure: Linux, Docker, GitHub Actions, Git, MongoDB, Redis, SQLite, CMake